



King Abdulaziz University

Faculty of Computing & Information Technology

Computer Science Department

Final Exam (April 27th, 2021)

Course: CPCS-223 (Analysis & Design of Algorithms)

Term: Spring 2021 (Semester 2, 1441/1442)

Points: 100

Time: 90 minutes

Student ID: _____

Student Name: _____

***Section: _____

CLO	SO	Question #

Instructor:

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MCQ (10 Points) **each right answer deserves 2 point**

SO-1

Height of the tree which has only one node is:

- (a) 0
- (b) 1
- (c) Can Not be calculated
- (d) None of above

Number of multiplications using “Strassen’s Matrix Multiplication”:

- (a) n^2
- (b) $n^{2.8}$
- (c) n^3
- (d) None of above

Fake-Coin problem using “decrease-by-a-constant factor” has running time

- (a) $\log_2 n$
- (b) $n \log_2 n$
- (c) n^2
- (d) None of above

The height of a complete binary tree is

- (a) The maximum path length from the root to a leaf node
- (b) Infinite
- (c) The maximum path length between two leaf nodes Linear
- (d) The number of leaf nodes Dynamically allocated
- (e) The number of nodes it contains None of these

If a graph is a “Directed Acyclic Graph” then is it not possible to apply “Topological Sorting”.

- (a) True
- (b) False

Question No. 1 (20 Points) *each correct iteration deserves 2 points and 4 points for the analysis* SO – 1

- Apply the following algorithm of Bubble Sort on the given data set and show the contents of the array after every iteration of outer loop.
- Furthermore, find the running time of the algorithm.

ALGORITHM *BubbleSort*($A[0..n - 1]$)
//Sorts a given array by bubble sort
//Input: An array $A[0..n - 1]$ of orderable elements
//Output: Array $A[0..n - 1]$ sorted in nondecreasing order
for $i \leftarrow 0$ **to** $n - 2$ **do**
 for $j \leftarrow 0$ **to** $n - 2 - i$ **do**
 if $A[j + 1] < A[j]$ **swap** $A[j]$ and $A[j + 1]$

	0	1	2	3	4	5	6	7	8
Initial	38	27	43	3	9	82	10	22	30
1 st	27	38	3	9	43	10	22	30	82
2 nd	27	3	9	38	10	22	30	43	82
3 rd	3	9	27	10	22	30	38	43	82
4 th	3	9	10	22	27	30	38	43	82
5 th	3	9	10	22	27	30	38	43	82
6 th	3	9	10	22	27	30	38	43	82
7 th	3	9	10	22	27	30	38	43	82
8 th	3	9	10	22	27	30	38	43	82
Final Values	3	9	10	22	27	30	38	43	82

Analysis:

$$\begin{aligned}
 C(n) &= \sum_{i=0}^{n-2} \sum_{j=0}^{n-2-i} 1 = \sum_{i=0}^{n-2} [(n-2-i) - 0 + 1] \\
 &= \sum_{i=0}^{n-2} (n-1-i) = \frac{(n-1)n}{2} \in \Theta(n^2).
 \end{aligned}$$

Question No. 2 (20 Points)**SO-2**

Using the given algorithm (Minimal Requirement Change), generate all permutations for;

S, T, A, R.

(Hint: start generating permutation with smallest character)

If $n = 1$ return 1;

Otherwise

1. generate recursively the list of all permutations of $12...n-1$
2. insert n into each of those permutations by starting with inserting n into $12...n-1$ by moving right to left
3. then switching direction for each new permutation

The permutations must be generated according to the algorithm. Random answers will not be accepted. Show all steps in generating the permutation.

Value	Permutations
A (1 pts)	A
R (1.5 for each row)	AR
	RA
S (2 for each row)	ARS, ASR, SAR
	SRA, RSA, RAS
T (2 for each row)	ARST, ARTS, ATRS, TARS
	ASRT, ASTR, ATSR, TASR
	SART, SATR, STAR, TSAR
	SRAT, SRTA, STRA, TSRA

	RSAT, RST ^A , RT ^S A, ^T RSA
	RAST, RAT ^S , RT ^A S, ^T RAS

Find the order of growth for solutions of the following recurrences.

a. $T(n) = 2T(n/4) + 1$

b. $T(n) = 2T(n/4) + \sqrt{n}$

c. $T(n) = 2T(n/2) + \Theta(n)$

Part – a:

$$T(n) = 2T(n/4) + 1$$

- $a = 2$
- $b = 4$
- $f(n) = 1 = n^0$
- $n^{\log_b a} = (n^{\log_4 2}) = (n^{\log_4 4^{1/2}}) = (n^{1/2}) = n^{1/2}$
 - $\varepsilon = 1/2$
 - Applying 1st case of Master Theorem Method
- $T(n) = \Theta(n^{1/2}) = \Theta(\sqrt{n})$

If $f(n) = O(n^{\log_b a - \varepsilon})$, for $\varepsilon > 0$ then $T(n) = \Theta(n^{\log_b a})$

Part – b:

$$T(n) = 2T(n/4) + \sqrt{n}$$

- $a = 2$
- $b = 4$
- $f(n) = \sqrt{n}$
- $n^{\log_b a} = (n^{\log_4 2}) = (n^{\log_4 4^{1/2}}) = (n^{1/2}) = n^{1/2}$
 - Applying 2nd case of Master Theorem Method
- $T(n) = \Theta(n^{1/2} \lg n) = \Theta(\sqrt{n} \lg n)$

If $f(n) = \Theta(n^{\log_b a})$, then $T(n) = \Theta(n^{\log_b a} \lg n)$

Part – c

$$T(n) = \underline{2T(n/2)} + \Theta(n)$$

$$T(n) = \underline{2T(n/2)} + c(n)$$

$$a = 2$$

$$b = 2$$

$$\underline{f(n)} = n$$

$$\underline{n}^{\log_b a} = n^{\log_2 2} = n$$

Applying 2nd case of Master Theorem Method

$$\text{If } f(n) = \Theta(n^{\log_b a}), \text{ then } T(n) = \Theta(n^{\log_b a} \lg n)$$

$$T(n) = \Theta(n \log_2 n)$$

Question No. 4 (25 Points) *each correct combination deserves 1 point and 1 point for the finding of the optimal solution* **SO - 1**

Find the minimum cost for the given data set using “Exhaustive Search” technique of solving “Assignment Problem” as discussed in class. You are required to show all calculations to find the solution.

	Job 1	Job 2	Job 3	Job 4
Person 1	9	2	7	8
Person 2	6	4	3	7
Person 3	5	8	1	8
Person 4	7	6	9	4

Solution:

$$C = \begin{bmatrix} 9 & 2 & 7 & 8 \\ 6 & 4 & 3 & 7 \\ 5 & 8 & 1 & 8 \\ 7 & 6 & 9 & 4 \end{bmatrix}$$

1, 2, 3, 4	cost = 9+4+1+4 = 18
1, 2, 4, 3	cost = 9+4+8+9 = 30
1, 3, 2, 4	cost = 9+3+8+4 = 24
1, 3, 4, 2	cost = 9+3+8+6 = 26
1, 4, 2, 3	cost = 9+7+8+9 = 33
1, 4, 3, 2	cost = 9+7+1+6 = 23

2, 1, 3, 4	cost = 2+6+1+4 = 13	3, 1, 2, 4	cost = 7+6+8+4 = 25
2, 1, 4, 3	cost = 2+6+8+9 = 25	3, 1, 4, 2	cost = 7+6+8+6 = 27
2, 3, 1, 4	cost = 2+3+5+4 = 14	3, 2, 1, 4	cost = 7+4+5+4 = 20
2, 3, 4, 1	cost = 2+3+8+7 = 20	3, 2, 4, 1	cost = 7+4+8+7 = 26
2, 4, 1, 3	cost = 2+7+5+9 = 23	3, 4, 1, 2	cost = 7+7+5+6 = 25
2, 4, 3, 1	cost = 2+7+1+7 = 17	3, 4, 2, 1	cost = 7+7+8+7 = 29
4, 1, 2, 3	cost = 8+6+8+9 = 31		
4, 1, 3, 2	cost = 8+6+1+6 = 21		
4, 2, 1, 3	cost = 8+4+5+9 = 26		
4, 2, 3, 1	cost = 8+4+1+7 = 20		
4, 3, 1, 2	cost = 8+3+5+6 = 22		
4, 3, 2, 1	cost = 8+3+8+7 = 26		

The optimal solution is: Person 1 to Job 2, Person 2 to Job 1, Person 3 to Job 3, and Person 4 to Job 4, with the total (minimal) cost of the assignment being 13.

Question No. 5 (10 Points)**SO - 1**

Write down the conditions for the following situations according to “The Game of Nim”, designed using “Variable Size Decrease Algorithm”. Note “**n**” are number of chips and “**m**” are number of chips a player can pick at the most.

S.No.	Situation	Condition
1	The player to move next is a “looser”	$n = 0$
2	The player to move next is a “looser”	$n = m + 1$
3	The player to move next is a “Winner”	$1 \leq n \leq m$
4	The player to move next is a “Winner”	$m+2 \leq n \leq 2m+1$
5	Winning Move	Leave multiple of (m+1) chips

Each condition deserves 2 points